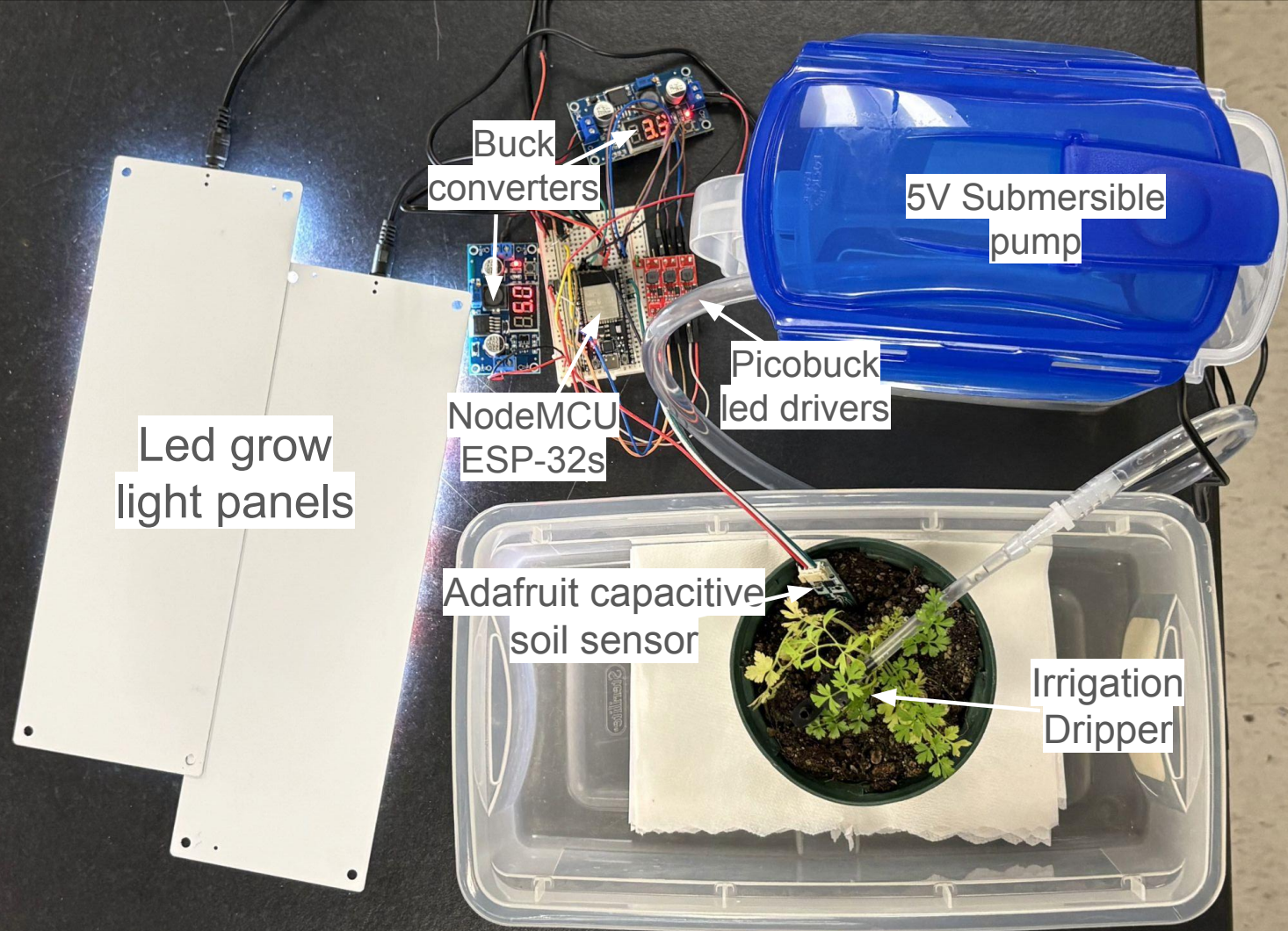


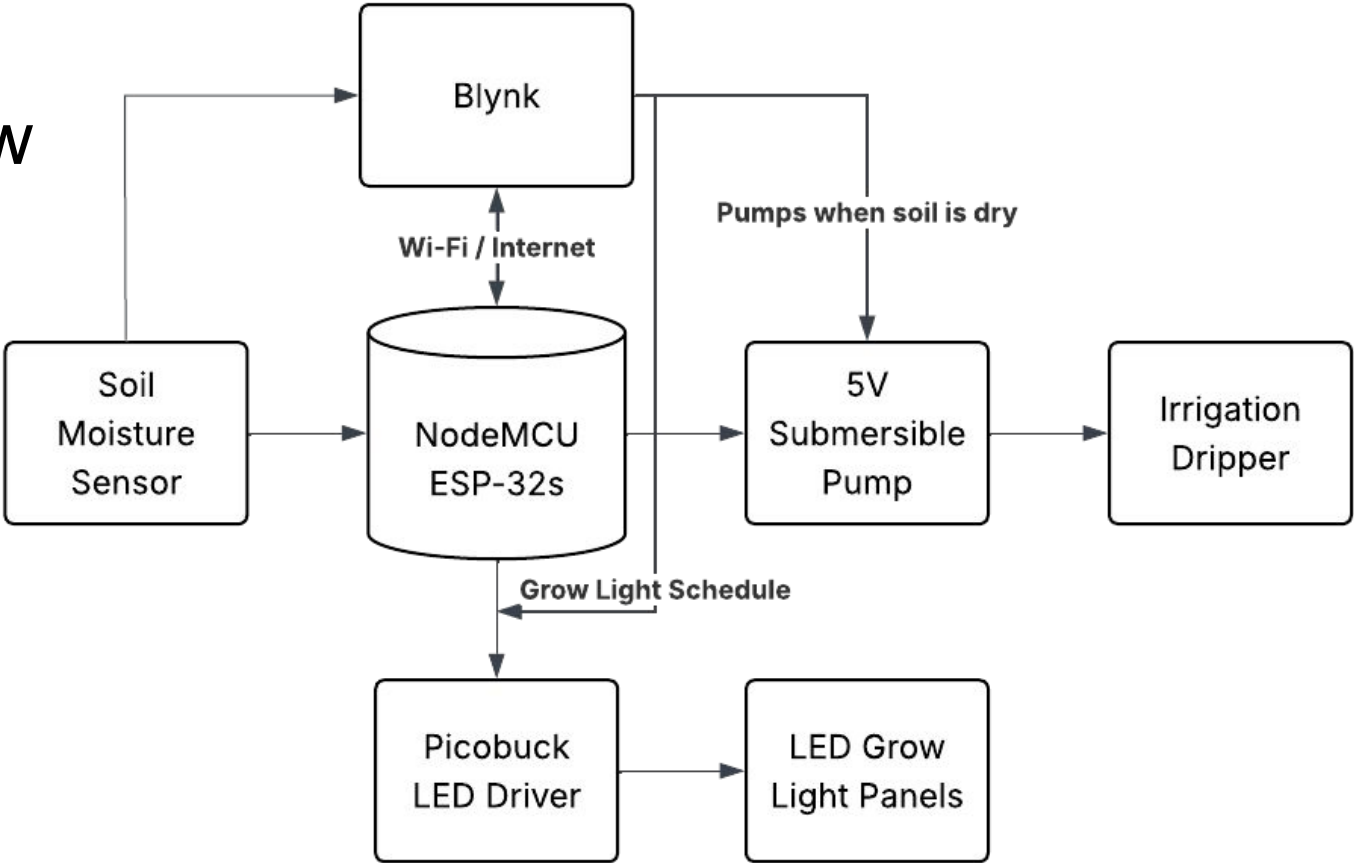
Fully Automated Planter Using ESP-32

Spencer Hughes

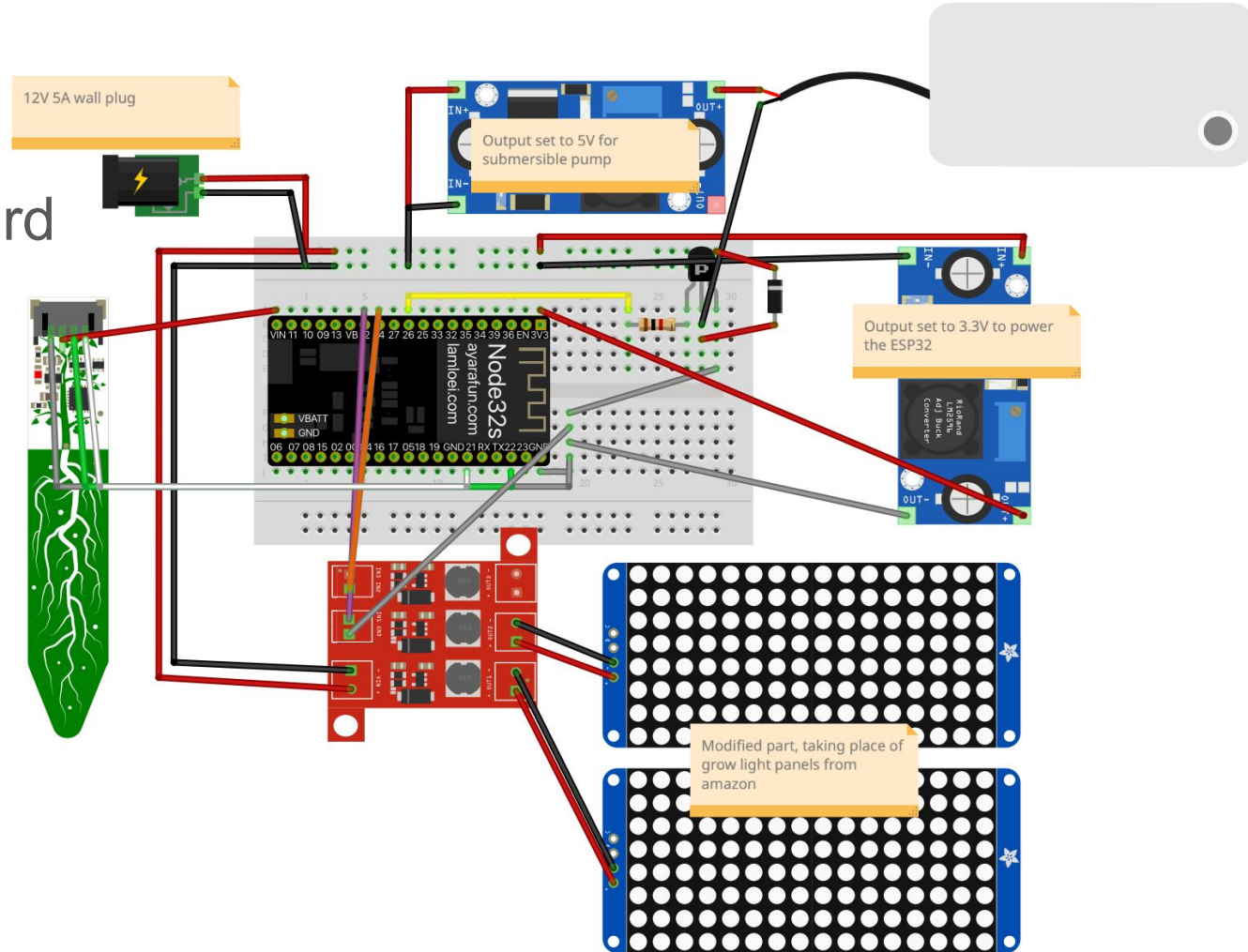
Assembly Diagram



Process Flow Diagram

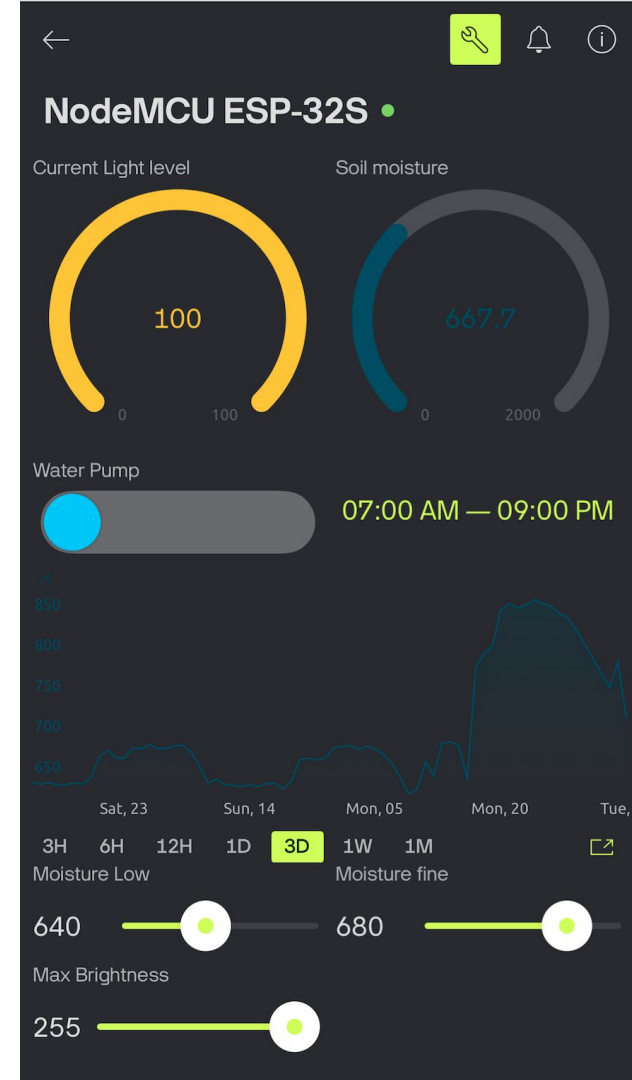


Fritzing Breadboard Diagram



Blynk Mobile App

- Blynk is an easy free way to put an arduino or Esp 32 online
- For information on how to use Blynk, refer to their documentation, they have templates to get started
- The app works using datastreams, which are baked into the code you're running on the board and are highly customizable for what you need



Physical Planter Assembly

- Flexpipe assembly was designed using 80/20 Idea Builder software
- Cam clamps were added to make the height of the grow lights adjustable found here
- Files: [design pdf](#) [design xml](#)



BOM for the “guts”

Link to the
[spreadsheet](#)

Part No.	Part Name	Supplier	Quantity	Units	Link	Notes
1	Grow Light	Amazon	1	4	link	
2	5V Submersible Pump	Amazon	1	1	link	
3	1/4 in Irrigation Tubing	Amazon	1	20ft	link	
4	Drip Irrigation Stake	Amazon	1	4	link	
5	Water Jug	Amazon	1	1	link	
6	NodeMCU ESP-32s	Amazon	1	4	link	
7	PicoBuck LED Driver	Sparkfun	1	1	link	Can also use FemtoBuck LED Drivers
8	LM2596 Buck Converter		2			Had in Dr. Krupczak's Lab
9	1N4001S Diode		1			Circuits Lab
10	TIP41C Transistor		1			Circuits Lab
11	1/2" x 1/4" Hose Reducer		1			Had in Senior design lab
12	STEMMA Soil Sensor	Adafruit	1	1	link	
13	STEMMA Cable	Adafruit	1	1	link	